PROJECT PROPOSAL: Parallel Image Processing Pipeline

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1. Theme Selection

We have selected **Theme A: Image Processing**. We will implement an image processing pipeline consisting of the following filters in order:

1. Gaussian Blur – reduces noise and smooths the image.
2. Median Filter – removes impulse noise (salt-and-pepper noise).
3. Sobel Edge Detection – computes image gradients to highlight edges.

2. Dataset Plan

We will use publicly available high-resolution images downloaded from the internet (e.g., Unsplash or similar free image sources) as our sources, and the sizes of our sources will be:

- **Small Input:** 512 × 512 (debugging)
- **Medium Input:** 1920 × 1080 (standard workload)
- **Large Input:** 4096 × 4096 (stress test for MPI communication and scalability)

3: Goals & Verification Plan

Goal:

To implement a serial baseline, an OpenMP parallel version, and an MPI distributed version of the image processing pipeline (Gaussian → Median → Sobel).

Verification:

We will compare the outputs of the OpenMP and MPI versions against the serial baseline using pixel-by-pixel comparison to ensure correctness. The outputs must match exactly or with a minor floating-point tolerance.

Performance Measurement:

We aim to achieve near-linear speedup using OpenMP on systems with no less than 8 cores and for large inputs (4096×4096), we expect significant performance improvements. We expect diminishing returns for MPI due to communication overhead, particularly during halo exchange between processes.